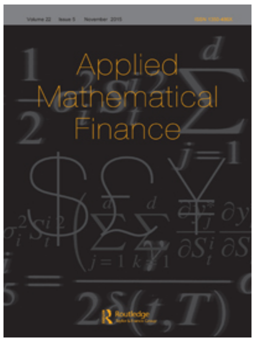

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Perpetual Exchange Options under Jump-Diffusion Dynamics

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ABSTRACT *This paper provides a pricing formula for perpetual exchange options, where the dynamics of the underlying assets are driven by jump-diffusion processes. It is an extension of Gerber and Shiu, and also Wong, who have priced perpetual exchange options under the pure-diffusion setting, and that of Gerber and Shiu, who have also considered perpetual options on single assets under jump-diffusion dynamics. It complements the results of Cheang and Chiarella, who derive a probabilistic representation of the American exchange option price under jump-diffusion dynamics.*

KEY WORDS: Perpetual American options, exchange options, compound Poisson processes, optimal stopping

1. Introduction

An exchange option allows the holder of the option to exchange one stock for another and is commonly used in foreign exchange markets, bond markets and stock markets. Margrabe (1978) derived a pricing formula for the European style exchange option in the Black-Scholes framework where the stock prices are modelled by correlated geometric Brownian motion processes. Recent extensions by Cheang, Chiarella, and Ziogas (2006) and Cheang and Chiarella (2011) extend Margrabe (1978) exchange option model to the jump-diffusion framework. A probabilistic representation of the American style exchange option under jump-diffusion dynamics is also derived in Cheang and Chiarella (2011).

The perpetual (American) exchange option under pure-diffusion dynamics was first considered by Gerber and Shiu (1996a, 1996b) who employed optimal stopping techniques, and was later generalized by Wong (2008). Liu and Liu (2009) consider the pricing of the perpetual option, again under pure-diffusion dynamics, from the perspective of solving a two-variable free boundary problem. The perpetual call and

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put options under pure-diffusion dynamics have been priced by Gerber and Shiu (1994), whereas the pricing of perpetual options on single stock under jump-diffusion is found in Gerber and Shiu (1994, 1998).

In this paper, we provide a pricing formula for perpetual exchange options, where the dynamics of the underlying assets are driven by jump-diffusion processes. The main result can be considered as an extension of Gerber and Shiu (1996a, 1996b), and Wong (2008), who have priced perpetual exchange options under the pure-diffusion setting, and that of Gerber and Shiu (1994, 1998), who have also considered perpetual options on single assets under jump-diffusion dynamics.

This paper is organized as follows. In Section 2, we specify the model dynamics of the two stocks under the jump-diffusion setting. In Section 3, we review existing results for European and American style exchange option pricing under the jump-diffusion dynamics framework. We introduce the perpetual exchange option in Section 4. In Section 5, the main result is stated and cast as an optimal stopping problem. Two examples are provided in Section 6. In Section 7, a counter-example with double-sided jumps is provided. We also consider a perpetual outperformance option in Section 8, then Section 9 concludes.

2. The Model Dynamics

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{Q})$ be a probability measure space. We assume that $\mathbb{Q}$ is an equivalent martingale measure that is already prescribed. The stock price dynamics are given by

$$\frac{dS_{i,t}}{S_{i,t-}} = (r - \zeta_i)dt + \sigma_i dW_{i,t} + \int_{\mathbb{R}} (e^{y_i} - 1)\hat{p}(dy_i, dt), \tag{1}$$

for $i = 1,\ 2$. In Equation (1), $W_{1,t}$ and $W_{2,t}$ are components of a bivariate correlated Brownian motion process which is adapted to the filtration, where $dW_{1,t}\, dW_{2,t} = \rho dt$, and ρ is the instantaneous correlation between the two Brownian motion components. It is also assumed that $|\rho| < 1$. The homogeneous Poisson counting measure for each stock is $p(dy_i, dt)$, adapted to the filtration, and each associated with the marked point process $(Y_{i,n}, N_{i,t})$, respectively. The marks $Y_{i,n}$ for each stock are independent and identically distributed, and independent of the two Poisson processes $N_{1,t}$ and $N_{2,t}$, as well as independent of the Brownian motion components. The intensity of each Poisson process $N_{i,t}$ is λ_i. Under $\mathbb{Q}$, the density of the identically distributed marks are $m_{\mathbb{Q}}(dy_i)$. We assume that $m_{\mathbb{Q}}(dy_i)$ is non-atomic. The compensated Poisson counting measure for each stock is

$$\hat{p}(dy_i, dt) = p(dy_i, dt) - \lambda_i m_{\mathbb{Q}}(dy_i)dt, \tag{2}$$

for $i = 1,\ 2$. The expected relative jump-size for each stock is $\kappa_i = \mathbb{E}_{\mathbb{Q}}[e^{Y_i} - 1]$. The risk-free interest rate is r and ζ_i is the continuously paying dividend rate for the $i$ith stock. It can be shown via Itô's Lemma for jump-diffusion processes (see e.g. Cont and Tankov [2004]), that the solution to Equation (1) is

$$S_{i,t} = S_{i,0}\ \exp\left[\left(r - \zeta_i - \frac{\sigma_i^2}{2} - \lambda_i\kappa_i\right)\ t + \sigma_i W_{i,t} + \sum_{n=1}^{N_{i,t}}\ Y_{i,n}\right], \tag{3}$$

for $i = 1, 2$. Under the chosen equivalent martingale measure $\mathbb{Q}$, the discounted stock yield processes $\{S_{i,t}e^{(\zeta_i - r)t}\}$ are martingales.

3. Option Pricing

We consider an exchange option which allows the holder of the option to exchange stock S_2 for S_1. In the case of a European exchange option, the payoff at maturity time T is

$$C_T^E(S_{1,T}, S_{2,T}) = (S_{1,T} - S_{2,T})^+. \tag{4}$$

In Cheang and Chiarella (2011), a pricing formula[1] is given for the European exchange option price

$$C_t^E(S_{1,t}, S_{2,t}) = e^{-r(T-t)}\mathbb{E}_{\mathbb{Q}}[(S_{1,T} - S_{2,T})^+|\mathcal{F}_t]. \tag{5}$$

A representation of the American option price $C_t^A(S_{1,t}, S_{2,t})$ (with finite maturity time T) in terms of the European price and an early exercise premium is also given in Cheang and Chiarella (2011).

We adopt the following notation C_{t-}^E to denote the pre-jump European option price $C_{t-}^E = C_{t-}^E(S_{1,t-}, S_{2,t-}) = C_t^E(S_{1,t-}, S_{2,t-})$. For ease of notation, we also denote the partial derivative with respect to time by ∂_t, the first-order partial derivative with respect to the ith stock price by ∂_i, and the second-order partial derivative with respect to the ith and jth stock prices by ∂_{ij}. Then the European exchange option price $C_t^E(S_{1,t}, S_{2,t})$ for our stock price model (Equation (1)) satisfies the integro-partial differential equation

$$\begin{aligned} &\partial_t C_{t-}^E + \mathcal{L}C_{t-}^E + \lambda_1\int_{\mathbb{R}}[C_t^E(S_{1,t-}e^{y_1}, S_{2,t-}) - C_{t-}^E]m_{\mathbb{Q}}(dy_1) \\ &\quad + \lambda_2\int_{\mathbb{R}}[C_t^E(S_{1,t-}, S_{2,t-}e^{y_2}) - C_{t-}^E]m_{\mathbb{Q}}(dy_2) = rC_{t-}, \end{aligned} \tag{6}$$

where

$$\mathcal{L} = \sum_{i=1}^{2}\left[(r - \zeta_i - \lambda_i\kappa_i)S_{i,t-}\partial_i + \frac{\sigma_i^2}{2}S_{i,t-}\partial_{ii}\right] + \rho\sigma_1\sigma_2 S_{1,t-}S_{2,t-}\partial_{12}.$$

It is shown in Cheang and Chiarella (2011) that even under jump-diffusion dynamics, the exchange option price $C_t(s_1, s_2)$ (whether European or American) is homogeneous in the stock price variables, that is

$$C_t(\gamma s_1, \gamma s_2) = \gamma C_t(s_1, s_2), \tag{7}$$

and as a consequence, it follows that

$$S_{1,t-}\partial_1 C_t(S_{1,t-}, S_{2,t-}) + S_{2,t-}\partial_2 C_t(S_{1,t-}, S_{2,t-}) = C_t(S_{1,t-}, S_{2,t-}). \tag{8}$$

Thus the European exchange option price $C_t^E(S_{1,t}, S_{2,t})$ also satisfies the integro-partial differential equation

$$\begin{aligned} &\partial_t C_{t-}^E + \mathcal{A}C_{t-}^E + \lambda_1 \int_{\mathbb{R}} [C_t^E(S_{1,t-}e^{y_1}, S_{2,t-}) - C_{t-}^E] m_{\mathbb{Q}}(dy_1) \\ &\quad + \lambda_2 \int_{\mathbb{R}} [C_t^E(S_{1,t-}, S_{2,t-}e^{y_2}) - C_{t-}^E] m_{\mathbb{Q}}(dy_2) = 0, \end{aligned} \tag{9}$$

where

$$\mathcal{A} = \sum_{i=1}^{2} \left[-(\zeta_i + \lambda_i \kappa_i) S_{i,t-}\partial_i + \frac{\sigma_i^2}{2} S_{i,t-}\partial_{ii} \right] + \rho\sigma_1\sigma_2 S_{1,t-}S_{2,t-}\partial_{12}.$$

For the American exchange option $C_t^A(S_{1,t}, S_{2,t})$ with a finite expiration time T, Cheang and Chiarella (2011) showed that there exists a time-dependent boundary b_t that separates the early exercise region $\{(S_{1,t}, S_{2,t}) : S_{1,t} \geq b_t S_{2,t}\}$ from the continuation region $\{(S_{1,t}, S_{2,t}) : S_{1,t} < b_t S_{2,t}\}$. In the continuation region, the American exchange option price $C_t^A(S_{1,t}, S_{2,t})$ satisfies the integro-partial differential equation

$$\begin{aligned} &\partial_t C_{t-}^A + \mathcal{A}C_{t-}^A + \lambda_1 \int_{\mathbb{R}} [C_t^A(S_{1,t-}e^{y_1}, S_{2,t-}) - C_{t-}^A] m_{\mathbb{Q}}(dy_1) \\ &\quad + \lambda_2 \int_{\mathbb{R}} [C_t^A(S_{1,t-}, S_{2,t-}e^{y_2}) - C_{t-}^A] m_{\mathbb{Q}}(dy_2) = 0, \end{aligned} \tag{10}$$

where C_{t-}^A denotes the pre-jump price of the American exchange option $C_{t-}^A = C_{t-}^A(S_{1,t-}, S_{2,t-}) = C_t^A(S_{1,t-}, S_{2,t-})$. The American exchange option price is also subject to the early exercise boundary condition (Equation (11)), together with the smooth pasting conditions (Equation (12)) and (Equation (13)) given below. The boundary and smooth-pasting conditions given the early exercise boundary b_t are

$$C_t^A(S_{1,t}, S_{2,t}) = S_{1,t} - S_{2,t} \quad \text{where} \quad S_{1,t} \geq b_t S_{2,t}, \tag{11}$$

$$\lim_{(s_1, s_2) \to (b_t s_2, s_2)} \frac{\partial C_t^A(s_1, s_2)}{\partial s_1} = 1, \tag{12}$$

$$\lim_{(s_1, s_2) \to (s_1, s_1/b_t)} \frac{\partial C_t^A(s_1, s_2)}{\partial s_2} = -1. \tag{13}$$

In addition, the condition

$$\frac{\partial C_t^A(s_1,s_2)}{\partial t}=0, \quad \text{where} \quad s_1>b_ts_2 \tag{14}$$

must also be satisfied. Furthermore, this is also an optional stopping problem where

$$C_t^A(S_{1,t},S_{2,t})=\sup_{t\le\tau\le T} e^{-r(\tau-t)}\mathbb{E}_{\mathbb{Q}}[(S_{1,\tau}-S_{2,\tau})^+|\mathcal{F}_t]. \tag{15}$$

4. The Perpetual Exchange Option

Now we turn our attention to the perpetual exchange option $C_t^P(S_{1,t},S_{2,t})$ under jump-diffusion dynamics. This is an optional stopping problem where

$$C_t^P(S_{1,t},S_{2,t})=\sup_{\tau\ge t} e^{-r(\tau-t)}\mathbb{E}_{\mathbb{Q}}[(S_{1,\tau}-S_{2,\tau})^+|\mathcal{F}_t]. \tag{16}$$

From the known properties of the perpetual call and put options under pure diffusion as well as jump-diffusion dynamics (Gerber and Shiu 1996a, 1996b), and that of a perpetual option on a single asset under jump-diffusion dynamics (Gerber and Shiu 1994, 1998), we can assume that our perpetual exchange option price is independent of time. Thus, we write

$$C_t^P(S_{1,t},S_{2,t})=C^P(S_{1,t},S_{2,t}). \tag{17}$$

Hence, it follows that the time partial derivative $\frac{\partial C^P(s_1,s_2)}{\partial t}$ is zero at all times t for $S_{1,t}=s_1$ and $S_{2,t}=s_2$.

Analogous to the case of the American exchange option under jump-diffusion dynamics in Section 3, there is an early exercise boundary that separates the early exercise region and continuation region. The key difference is that the early exercise boundary is independent of time. The early exercise boundary b separates the early exercise region $\{(S_{1,t},S_{2,t}):S_{1,t}\ge bS_{2,t}\}$ from the continuation region $\{(S_{1,t},S_{2,t}):S_{1,t}<bS_{2,t}\}$. Let the pre-jump stock prices be $S_{1,t-}=s_1$ and $S_{2,t-}=s_2$, and the pre-jump perpetual exchange option price be $C_{t-}^P=C^P(s_1,s_2)$. Then in the continuation region, the perpetual exchange option satisfies the partial differential equation

$$\begin{aligned}\mathcal{A}C_{t-}^P&+\lambda_1\int_{\mathbb{R}}[C_t^P(s_1e^{y_1},s_2)-C_{t-}^P]m_{\mathbb{Q}}(dy_1)\\&+\lambda_2\int_{\mathbb{R}}[C_t^P(s_1,s_2e^{y_2})-C_{t-}^P]m_{\mathbb{Q}}(dy_2)=0.\end{aligned} \tag{18}$$

Figure 1 demonstrates the continuation and stopping regions, and early exercise boundary for the perpetual American exchange option under consideration. In contrast to the American exchange option under geometric Brownian motion (Broadie

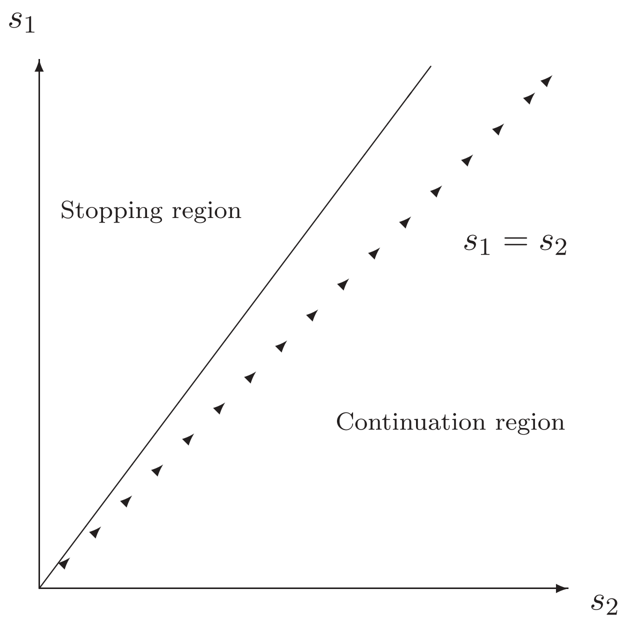

Figure 1. Continuation and stopping regions for the perpetual American exchange option, at any given time t. The early exercise boundary is the line $s_1 = bs_2$.

and Detemple 1997) and the American exchange option under jump-diffusion (Cheang and Chiarella 2011), the early exercise boundary $s_1 = bs_2$ in Figure 1 is independent of time.

The perpetual exchange option price is also subject to the early exercise boundary condition (Equation (19)), together with the smooth pasting conditions (Equations (20) and (21)) given below. The boundary and smooth-pasting conditions given the early exercise boundary b are

$$C^P(s_1, s_2) = s_1 - s_2 \quad \text{where} \quad s_1 \geq bs_2, \tag{19}$$

$$\lim_{(s_1,s_2)\to(bs_2,s_2)} \frac{\partial C^P(s_1, s_2)}{\partial s_1} = 1, \tag{20}$$

$$\lim_{(s_1,s_2)\to(s_1,s_1/b)} \frac{\partial C^P(s_1, s_2)}{\partial s_2} = -1. \tag{21}$$

Unlike the case of American exchange options with finite expiration times, there exists a closed-form formula for the perpetual exchange option in the continuation region.

5. An Optimal Stopping Problem

In the case of single stock perpetual call and perpetual put options, Gerber and Shiu (1994, 1998) showed that optimal solutions exist for perpetual calls if the stock is skip-free upwards, whereas those of perpetual puts exist if the stock is skip-free downwards. Since the payoff of the exchange option takes the form $(S_{1,t} - S_{2,t})^+$, we impose the condition that stock S_1 is skip-free upwards and stock S_2 is skip-free downwards.

As we are working under an equivalent martingale measure $\mathbb{Q}$, the discounted stock yield processes $\{S_{1,t}e^{(\zeta_1-r)t}\}$ and $\{S_{2,t}e^{(\zeta_2-r)t}\}$ are martingales. Since the perpetual exchange option price $C^P(s_1, s_2)$ is also homogeneous in the two stock variables, we seek solutions of the form $C^P(s_1, s_2) = Ks_1^\theta s_2^{1-\theta}$ in the continuation region. Indeed we see later in this section that in the continuation region, $C^P(s_1, s_2)$ takes the form $C(s_1, s_2, b) = K(b, \theta)s_1^\theta s_2^{1-\theta}$ for some optimal b that optimizes the value of $K(b, \theta)s_1^\theta s_2^{1-\theta}$ and this parameter b is early exercise boundary. Thus, following Gerber and Shiu (1996a, 1996b), we seek the following discounted process $\{e^{-rt}S_{1,t}^\theta S_{2,t}^{1-\theta}\}$ which will be a martingale under $\mathbb{Q}$ for some $\theta > 1$. We now state our main result.

Theorem 1. Suppose we have two stocks S_1 and S_2 driven by the dynamics (Equation (1)) under an equivalent martingale measure $\mathbb{Q}$. If there is a $\theta > 1$ that satisfies

$$\begin{aligned} &\frac{1}{2}\theta(1-\theta)v^2 + \theta\zeta_1 + (1-\theta)\zeta_2 - \lambda_1[m_{\mathbb{Q},Y_1}(\theta) - 1] \\ &\quad - \lambda_2[m_{\mathbb{Q},Y_2}(1-\theta) - 1] + \theta\lambda_1\kappa_1 + (1-\theta)\lambda_2\kappa_2 = 0, \end{aligned} \tag{22}$$

where $v^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, then $\{e^{-rt}S_{1,t}^\theta S_{2,t}^{1-\theta}\}$ is a martingale under $\mathbb{Q}$, and the price of the perpetual exchange option $C^P(S_1, S_2)$ at any time $t > 0$ to exchange stock S_2 for S_1 is given by

$$C^P(S_1, S_2) = \begin{cases} \left(\frac{S_1}{\theta}\right)^\theta / \left(\frac{S_2}{\theta-1}\right)^{\theta-1} & \text{if} \quad S_1 < \frac{\theta}{\theta-1}S_2, \\ S_1 - S_2 & \text{if} \quad S_1 \geq \frac{\theta}{\theta-1}S_2. \end{cases} \tag{23}$$

Proof. Since the price of the perpetual exchange option has to be independent of time, in place of the optimal stopping problem (Equation (16)), we can simply consider the problem

$$C^P(S_1, S_2) = \sup_{\tau\geq 0} e^{-r\tau}\,\mathbb{E}_\mathbb{Q}[(S_{1,\tau} - S_{2,\tau})^+] = \sup_{\tau\geq 0}\,\mathbb{E}_\mathbb{Q}[e^{-r\tau}(S_{1,\tau} - S_{2,\tau})^+\mathbb{1}_{\{\tau<\infty\}}] \tag{24}$$

instead. Let

$$\tau_b = \inf\{\tau \geq 0 | S_{1,\tau} \geq bS_{2\tau}\}.$$

Then

$$\begin{aligned} C^P(bS_2, S_2) &= \mathbb{E}_{\mathbb{Q}}[e^{-r\tau_b}(bS_{2,\tau_b} - S_{2,\tau_b})^+ \mathbb{1}_{\{\tau_b<\infty\}}] \\ &= (b-1)\mathbb{E}_{\mathbb{Q}}\left[e^{-r\tau_b} S_{2,\tau_b} \mathbb{1}_{\{\tau_b<\infty\}}\right]. \end{aligned} \tag{25}$$

Assuming that we have already found a $\theta > 1$ for which $\{e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\}$ is a martingale under $\mathbb{Q}$, the stopped process $\{e^{-rt\wedge\tau_b}S_{1,t\wedge\tau_b}^{\theta}S_{2,t\wedge\tau_b}^{1-\theta}\}$ is also a martingale. Hence,

$$\begin{aligned} S_{1,0}^{\theta}S_{2,0}^{1-\theta} &= \mathbb{E}_{\mathbb{Q}}\left[e^{-r\tau_b}S_{1,\tau_b}^{\theta}S_{2,\tau_b}^{1-\theta}\mathbb{1}_{\{\tau_b \le t\}}\right] \\ &\quad + \mathbb{E}_{\mathbb{Q}}\left[e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\mathbb{1}_{\{\tau_b > t\}}\right]. \end{aligned} \tag{26}$$

If $\tau_b > t$, then $S_{1,t} < bS_{2,t}$, and since $\theta > 0$, we can bound

$$S_{1,t}^{\theta}S_{2,t}^{1-\theta}\mathbb{1}_{\{\tau_b > t\}} < b^{\theta}S_{2,t}.$$

For the last term $\mathbb{E}_{\mathbb{Q}}\left[e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\mathbb{1}_{\{\tau_b > t\}}\right]$ in (Equation (26)), we have

$$\mathbb{E}_{\mathbb{Q}}\left[e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\mathbb{1}_{\{\tau_b > t\}}\right] \le b^{\theta}S_{2,0}e^{-\zeta_2 t}, \tag{27}$$

since $\{S_{2,t}e^{(\zeta_2-r)t}\}$ is a $\mathbb{Q}$-martingale. The bound $b^{\theta}S_{2,0}e^{-\zeta_2 t}$ goes to zero as $t \to \infty$, and thus (Equation (26)) becomes

$$S_1^{\theta}S_2^{1-\theta} = S_{1,0}^{\theta}S_{2,0}^{1-\theta} = b^{\theta}\mathbb{E}_{\mathbb{Q}}\left[e^{-r\tau_b}S_{2,\tau_b}\mathbb{1}_{\{\tau_b<\infty\}}\right] . \tag{28}$$

Substituting (Equation (28)) into (Equation ((25)), we have in the continuation region

$$C^P(S_1, S_2) = \max_{b>1}\ (b-1)b^{-\theta}S_1^{\theta}S_2^{1-\theta}. \tag{29}$$

The expression above is maximized at $b = \frac{\theta}{\theta-1}$ and thus we obtain Equation (23) if a $\theta > 1$ exists such that $\{e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\}$ is a martingale under $\mathbb{Q}$.

Now we discuss how to find θ. For $\{e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\}$ to be a martingale under $\mathbb{Q}$, we require

$$e^{-rt}\mathbb{E}_{\mathbb{Q}}[S_{1,t}^{\theta}S_{2,t}^{1-\theta}] = S_{1,0}^{\theta}S_{2,0}^{1-\theta}, \tag{30}$$

which leads to

$$e^{-rt}\,\mathbb{E}_{\mathbb{Q}}\left[\exp\left(\sum_{i=1}^{i=2}\ \theta_i\left[\left(r-\zeta_i-\frac{\sigma_i^2}{2}-\lambda_i\kappa_i\right)\ t+\sigma_i W_{i,t}+\sum_{n=1}^{N_{i,t}}\ Y_{i,n}\right]\right)\right]=1, \tag{31}$$

where $\theta_1 = \theta$ is to be determined and $\theta_2 = 1-\theta$. Hence, as a consequence of Equation (31), the perpetual exchange option price (Equation (23)) exists if there is a solution for θ that satisfies Equation (22).□

It can be easily verified that the perpetual exchange option price (Equation (23)) in Theorem 1 satisfies the boundary condition (Equation (19)) and the smooth pasting conditions (Equations (20) and (21)).

6. Some Examples

We provide two examples to show how the theorem works.

Example 1. Suppose stock S_1 has downwards jumps independently and identically distributed as negative exponentials, that is

$$f_{Y_{1,n}}(y)=\begin{cases}\alpha_1 e^{\alpha_1 y}, & \text{for } y<0,\\ 0 & \text{otherwise;}\end{cases} \tag{32}$$

and stock S_2 has upwards jumps independently and identically distributed as exponentials, that is

$$f_{Y_{2,n}}(y)=\begin{cases}\alpha_2 e^{-\alpha_2 y}, & \text{for } y>0,\\ 0 & \text{otherwise.}\end{cases} \tag{33}$$

The corresponding moment-generating functions of $Y_{1,n}$ and $Y_{2,n}$ are $m_{\mathbb{Q},Y_1}(u)=\frac{\alpha_1}{\alpha_1+u}$ and $m_{\mathbb{Q},Y_2}(u)=\frac{\alpha_2}{\alpha_2-u}$, respectively. Then the parameter $\theta>1$ for the perpetual exchange option price (Equation (23)) needs to satisfy

$$\frac{1}{2}\theta(1-\theta)v^2+\theta\zeta_1+(1-\theta)\zeta_2+\frac{\lambda_1\theta}{\alpha_1+\theta}+\frac{\lambda_2(\theta-1)}{\alpha_1-1+\theta}-\frac{\lambda_1\theta}{\alpha_1+1}+\frac{\lambda_2(1-\theta)}{\alpha_2-1}=0. \tag{34}$$

Suppose our parameters are given in the four cases in Table 1, then the solutions to θ in Equation (34) are shown in Table 2. We note in Table 2 that for each case there is only one value of $\theta=\theta_1$ where $\theta>1$. Hence, the perpetual option price in each case is given by Equation (23) for this particular value of θ, since all the other values of θ that are roots of Equation (34) are less than 1.

Example 2. We consider one-sided gamma random variables for this example. Suppose stock S_1 has downwards jumps independently and identically distributed as negative gamma, that is

Table 1. The parameters when the jumps are exponentially distributed.

Parameters	Case 1	Case 2	Case 3	Case 4
σ_1	0.15	0.15	0.1	0.1
ζ_1	0.02	0.02	0.015	0.015
σ_2	0.1	0.1	0.15	0.15
ζ_2	0.015	0.015	0.02	0.02
λ_1	0.2	0.1	0.2	0.1
λ_2	0.1	0.2	0.1	0.2
α_1	1/0.06	1/0.03	1/0.06	1/0.03
α_2	1/0.03	1/0.03	1/0.03	1/0.03
ρ	–0.8	0.8	–0.8	0.8

Table 2. The computed θ when jumps are exponentially distributed.

θ	Case 1	Case 2	Case 3	Case 4
θ_1	1.51	3.16	1.34	2.05
θ_2	−0.34	−1.05	−0.51	−2.15
θ_3	−17.06	−32.89	−17.07	−32.90
θ_4	−32.45	−34.85	−32.45	−34.99

$$f_{Y_{1,n}}(y) = \begin{cases} \frac{y^{\alpha_1 - 1} e^{y/\beta_1}}{\beta_1^{\alpha_1}\Gamma(\alpha_1)}, & \text{for } y < 0, \\ 0 & \text{otherwise;} \end{cases} \tag{35}$$

and stock S_2 has upwards jumps independently and identically distributed as gamma, that is

$$f_{Y_{2,n}}(y) = \begin{cases} \frac{y^{\alpha_2 - 1} e^{-y/\beta_2}}{\beta_2^{\alpha_2}\Gamma(\alpha_2)}, & \text{for } y > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{36}$$

The corresponding moment-generating functions of $Y_{1,n}$ and $Y_{2,n}$ are $m_{\mathbb{Q},Y_1}(u) = \frac{1}{(1+\beta_1 u)^{\alpha_1}}$ and $m_{\mathbb{Q},Y_2}(u) = \frac{1}{(1-\beta_2 u)^{\alpha_2}}$, respectively. Then the parameter $\theta > 1$ for the perpetual exchange option price (Equation (23)) needs to satisfy Equation (22) with the corresponding moment-generating functions for the negative gamma and gamma. The parameters for four cases are given in Table 3.

The values of θ for each case is given in Table 4. For each case, the option price is given by (Equation (23)) for $\theta = \theta_1$ since θ_1 is the only value of θ greater than 1.

7. A Counter-Example

We provide a counter-example to show that if we relax the assumptions on stock S_1 being skip-free upwards and stock S_2 being skip-free downwards, then Theorem 1 does not hold in general. Suppose we let $Y_{1,n}$ be normally distributed as $N(\alpha_1, \delta_1^2)$ and

Table 3. The parameter list when jumps are gamma distributed.

Parameters	Case 1	Case 2	Case 3	Case 4
σ_1	0.15	0.15	0.1	0.1
ζ_1	0.02	0.02	0.015	0.015
σ_2	0.1	0.1	0.15	0.15
ζ_2	0.015	0.015	0.02	0.02
λ_1	0.2	0.1	0.2	0.1
λ_2	0.1	0.2	0.1	0.2
ρ	− 0.8	0.8	− 0.8	0.8
α_1	2	2	2	2
α_2	2	2	2	2
β_1	0.015	0.030	0.030	0.015
β_2	0.015	0.015	0.015	0.015

Table 4. The computed θ when the jumps are gamma distributed.

θ	Case 1	Case 2	Case 3	Case 4
θ_1	1.52	3.18	1.34	2.06
θ_2	−0.34	−1.06	−0.52	−2.18
θ_3	−65.72 − 1.89i	−67.04 − 8.25i	−65.72 − 1.90i	−67.09 − 8.3i
θ_4	−65.72 + 1.89i	−67.04 + 8.25i	−65.72 + 1.90i	−67.09 + 8.3i
θ_5	2.60 − 33.53i	−66.33 − 0.47i	−33.54 − 2.62i	−66.34 − 0.47i
θ_6	2.60 + 33.53i	−66.33 − 0.47i	−33.54 + 2.62i	−66.34 − 0.47i

$Y_{2,n}$ be normally distributed as $N(\alpha_2, \delta_2^2)$. The corresponding moment-generating functions are then $m_{\mathbb{Q},Y_i}(u) = e^{u\alpha_i + 0.5u^2\delta_i^2}$ for $i = 1,\ 2$. We show in Table 5 that there are some parameters of the diffusion and the jump components where no feasible solution for θ in Equation (22) exists in order for us to apply the perpetual option price (Equation (23)). For the parameters of the two cases in Table 5, the two values of θ are both less than 1.

8. A Perpetual Outperformance Option

An important example of an exchange option is the outperformance option. The holder of the outperformance option gains the amount by which one asset outperforms some benchmark, for example an index. Here we consider the situation in which the holder of a perpetual outperformance option gains if a stock subject to idiosyncratic jump-risk outperforms a particular well-diversified index at the time that he/she chooses to exercise the option.

We consider a dividend-paying stock

$$\frac{dS_{1,t}}{S_{1,t-}} = (r - \zeta_1)dt + \sigma_1 dW_{1,t} + \int_{\mathbb{R}} (e^{y_1} - 1)\hat{p}(dy_1, dt), \tag{37}$$

Table 5. The parameters and θ when jumps are normally distributed.

	Case 1	Case 2
σ_1	0.1	0.1
ζ_1	0.01	0.015
λ_1	0.4	0.5
α_1	0.02	0.02
δ_1	0.06	0.01
σ_2	0.15	0.1
ζ_2	0.03	0.02
λ_2	0.1	0.2
α_2	0.01	0.02
δ_2	0.01	0.01
ρ	− 0.2	− 0.8
θ_1	0.86	0.87
θ_2	− 1.74	− 1.26

under a prescribed equivalent martingale measure $\mathbb{Q}$, with continuously paying dividend rate ζ_1, and a well-diversified index

$$dS_{2,t} = (r - \zeta_2)S_{2,t}dt + \sigma_2 S_{2,t}dW_{2,t}. \tag{38}$$

The index also has a cost of carry ζ_2 due to the dividends of the stocks that make up the index. The other assumption is that the Brownian motion process driving the stock and the Brownian motion process driving the index are correlated with $dW_{1,t}\, dW_{2,t} = \rho dt$. Since the index is assumed to be truly diversified, it is not subjected to idiosyncratic risk. However, the stock is subjected to idiosyncratic risk with jump-sizes Y_1. The intensity of the jump arrivals is λ_1 in the market measure $\mathbb{Q}$ and the expected relative jump-size is

$$\kappa_1 = \mathbb{E}_{\mathbb{Q}}[e^{Y_1} - 1] = \int_{\mathbb{R}} [e^{y_1} - 1] m_{\mathbb{Q}}(dy_1).$$

As it is in the case of the perpetual exchange option, the payoff of the outperformance option when exercised at time t is $(S_{1,t} - S_{2,t})^+$. We also impose the condition that stock S_1 is skip-free upwards. A similar analysis to that of the perpetual exchange option can be invoked to show that the price of the perpetual outperformance option $C^P(S_1, S_2)$ has to be independent of time. Furthermore, since we are working under an equivalent martingale measure $\mathbb{Q}$, the discounted stock yield process $\{ S_{1,t}e^{(\zeta_1 - r)t} \}$ and the discounted index process $\{ S_{2,t}e^{(\zeta_2 - r)t} \}$ are martingales. The price of the perpetual outperformance option $C^P(S_1, S_2)$ is stated in the following corollary.

Corollary 1. Suppose we have a stock S_1 and a well-diversified index S_2 driven by the dynamics (Equation (37)) and (Equation (38)), respectively, under an equivalent martingale measure $\mathbb{Q}$. If there is a $\theta > 1$ that satisfies

$$\frac{1}{2}\theta(1-\theta)v^2+\theta\zeta_1+(1-\theta)\zeta_2-\lambda_1[m_{\mathbb{Q},Y_1}(\theta)-1]+\theta\lambda_1\kappa_1=0, \tag{39}$$

where $v^2=\sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2$, then $\{e^{-rt}S_{1,t}^{\theta}S_{2,t}^{1-\theta}\}$ is a martingale under $\mathbb{Q}$, and the price of the perpetual outperformance option $C^P(S_1,S_2)$ at any time $t>0$ is given by

$$C^P(S_1,S_2)=\begin{cases}\left(\frac{S_1}{\theta}\right)^{\theta}/\left(\frac{S_2}{\theta-1}\right)^{\theta-1} & \text{if} \quad S_1<\frac{\theta}{\theta-1}S_2,\\ S_1-S_2 & \text{if} \quad S_1\geq\frac{\theta}{\theta-1}S_2.\end{cases} \tag{40}$$

9. Conclusion

We provide a pricing formula for perpetual exchange options under jump-diffusion dynamics in this paper. The pricing formula applies when each stock has single-sided jumps going in opposite directions towards each other. Two examples are provided to show how a parameter for the early exercise boundary and the option price is calculated. A counter-example is provided to show that the result does not apply if the jumps for both stocks are double-sided.

Note

[1]See also Caldana et al. (2015) for the correction to a formula in Cheang and Chiarella (2011).

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